

JEE EXERCISES

Exercise J-A

1. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation $(\sqrt[3]{1+a}-1)x^2 + (\sqrt{1+a}-1)x + (\sqrt[6]{1+a}-1) = 0$ Where $a > -1$, then $\lim_{a \rightarrow 0^+} \alpha(a)$ and $\lim_{a \rightarrow 0^+} \beta(a)$ are : [JEE-2012]

माना समीकरण $(\sqrt[3]{1+a}-1)x^2 + (\sqrt{1+a}-1)x + (\sqrt[6]{1+a}-1) = 0$, जहाँ $a > -1$, के मूल $\alpha(a)$ तथा $\beta(a)$ हैं, तो $\lim_{a \rightarrow 0^+} \alpha(a)$ तथा $\lim_{a \rightarrow 0^+} \beta(a)$ का मान होगा : [JEE-2012]

- (A) $-\frac{5}{2}$ & 1 (B) $-\frac{1}{2}$ & -1 (C) $-\frac{7}{2}$ & 2 (D) $-\frac{9}{2}$ & 3

Ans. B

Sol. $\left(\left(1 + \frac{a}{3} \right) - 1 \right) x^2 + \left(\left(1 + \frac{a}{2} \right) - 1 \right) x + \left(1 + \frac{a}{6} - 1 \right) = 0$

$$a \left(\frac{x^2}{3} + \frac{x}{2} + \frac{1}{6} \right) = 0 \Rightarrow 2x^2 + 3x + 1 = 0$$

$$\Rightarrow x = -\frac{1}{2}, -1$$

$$\Rightarrow \lim_{a \rightarrow 0^+} \alpha(a) \text{ and } \lim_{a \rightarrow 0^+} \beta(a) \text{ are } -\frac{1}{2} \text{ and } -1$$

2. If $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$ then : [JEE-2012]

यदि $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$ तो : [JEE-2012]

- (A) $a = 1, b = 4$ (B) $a = 1, b = -4$ (C) $a = 2, b = -3$ (D) $a = 2, b = 3$

Ans. B

Sol. $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$

$$\lim_{x \rightarrow \infty} \left(\frac{x^2(1-a) + x(1-a-b) + (1-b)}{x+1} \right) = 4$$

Limit is finite

$$\text{It exists when } 1 - a = 0 \Rightarrow a = 1$$

$$\text{then } \lim_{x \rightarrow \infty} \left(\frac{1 - a - b + \frac{1-b}{x}}{1 + \frac{1}{x}} \right) = 4$$

$$\therefore 1 - a - b = 4 \Rightarrow b = -4$$

3. The largest value of the nonnegative integer a for which $\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$ is :
[JEE(Adv.)2014]

a का अधिकतम अऋणात्मक पूर्णांक मान ज्ञात कीजिये जिसके लिये $\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$ होगा :
[JEE(Adv.)2014]

Ans. Zero

Sol. $\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$

$$\Rightarrow \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{x+\sqrt{x}} = \frac{1}{4}$$

Hence $\left\{ \frac{-ax + \sin(x-1) + a}{(x-1) + \sin(x-1)} \right\}^{1+\sqrt{x}} = \frac{1}{4}$

put $x = 1 + h$,

$$\lim_{h \rightarrow 0} \left\{ \frac{-ah + \sinh}{h + \sinh} \right\}^{1+\sqrt{1+h}} = \frac{1}{4}$$

or $\frac{-a+1}{2} = \frac{1}{2}$ or $-\frac{1}{2} \Rightarrow a = 0$ or 2

But at $a = 2$, $\frac{-ah + \sinh}{h + \sinh}$ tends to negative value

So correct Answer is $a = 0$

However $a = 2$ may be accepted if this is not considered

4. Let $f(x) = \sin \left(\frac{\pi}{6} \sin \left(\frac{\pi}{2} \sin x \right) \right)$ for all $x \in \mathbb{R}$ and $g(x) = \frac{\pi}{2} \sin x$ for all $x \in \mathbb{R}$. Let $(f \circ g)(x)$ denote $f(g(x))$ and $(g \circ f)(x)$ denote $g(f(x))$. Then which of the following is (are) true?
[JEE(Adv.)2015]

(A) Range of f is $\left[-\frac{1}{2}, \frac{1}{2} \right]$

(B) Range of $f \circ g$ is $\left[-\frac{1}{2}, \frac{1}{2} \right]$

(C) $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \frac{\pi}{6}$

(D) There is an $x \in \mathbb{R}$ such that $(g \circ f)(x) = 1$

माना कि सभी $x \in \mathbb{R}$ के लिए $f(x) = \sin\left(\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)\right)$ और सभी $x \in \mathbb{R}$ के लिए $g(x) = \frac{\pi}{2} \sin x$ है। माना कि $(f \circ g)(x)$ और $(g \circ f)(x)$ क्रमशः $f(g(x))$ और $g(f(x))$ को दर्शाते हैं, तब निम्नलिखित में से कौन सही है? **[JEE(Adv.)2015]**

(A) f का परिसर $\left[-\frac{1}{2}, \frac{1}{2}\right]$ है

(B) $f \circ g$ का परिसर $\left[-\frac{1}{2}, \frac{1}{2}\right]$ है

(C) $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \frac{\pi}{6}$

(D) $\mathbb{R}$ में एक x ऐसा है जिसके लिए $(g \circ f)(x) = 1$

Ans. ABC

Sol. $f(x) = \sin\left(\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)\right)$

Let $\frac{\pi}{2} \sin x = \theta \quad \because \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\therefore f(x) = \sin\left(\frac{\pi}{6} \sin \theta\right)$

Let $\frac{\pi}{6} \sin \theta = \phi \quad \therefore \phi \in \left[-\frac{\pi}{6}, \frac{\pi}{6}\right]$

$\therefore f(x) = \sin \phi \in \left[-\frac{1}{2}, \frac{1}{2}\right] \quad \therefore \textbf{(A)}$

Now $f \circ g(x) = \sin\left(\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin\left(\frac{\pi}{2} \sin x\right)\right)\right)$

Clearly, range of $f \circ g$ is also $\left[-\frac{1}{2}, \frac{1}{2}\right] \quad \therefore \textbf{(B)}$

Now, $\lim_{x \rightarrow 0} \frac{\sin\left(\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)\right)}{\frac{\pi}{2} \sin x}$

$= \lim_{x \rightarrow 0} \frac{2 \sin\left(\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)\right)}{\pi \frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)} \times \frac{\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)}{\frac{\sin x}{x} \times x}$

$= \lim_{x \rightarrow 0} \frac{2}{\pi} \times \frac{\pi}{6} \times \frac{\sin\left(\frac{\pi}{2} \sin x\right)}{\frac{\pi}{2} \sin x} \times \frac{\frac{\pi}{2} \sin x}{x}$

$$\frac{1}{3} \times \frac{\pi}{2} = \frac{\pi}{6} \quad \therefore \quad \text{(C)}$$

Now, $\sin f(x) = 1$

$$\Rightarrow \frac{\pi}{2} \sin \left(\frac{\pi}{2} \sin \left(\frac{\pi}{2} \sin x \right) \right) = 1$$

$$\Rightarrow \sin \left(\frac{\pi}{6} \sin \left(\frac{\pi}{2} \sin x \right) \right) = \frac{2}{\lambda} \cong \frac{2}{3.14} > \frac{1}{2} \quad \therefore \quad \text{(D)}$$

5. Let m and n be two positive integers greater than 1. If $\lim_{\alpha \rightarrow 0} \left(\frac{e^{\cos(\alpha^n)} - e}{\alpha^m} \right) = -\left(\frac{e}{2}\right)$. Then the value of $\frac{m}{n}$ is :

[JEE(Adv.)2015]

माना कि दो धनात्मक पूर्णांक m तथा n एक (1) से बड़े हैं (greater than 1) । यदि $\lim_{\alpha \rightarrow 0} \left(\frac{e^{\cos(\alpha^n)} - e}{\alpha^m} \right) = -\left(\frac{e}{2}\right)$ तब m/n का मान है:

[JEE(Adv.)2015]

Ans. 2

Sol. $\therefore m \geq 2$ and और $n \geq 2$

$$\begin{aligned} &= \lim_{a \rightarrow 0} \frac{e(e^{\cos(a^n)-1} - 1)}{(\cos(a^n) - 1)} \times \left(\frac{\cos(a^n) - 1}{(a^n)^2} \right) \frac{a^{2n}}{a^m} \\ &= e \times \lim_{a \rightarrow 0} \left(\frac{e^{\cos(a^n)-1} - 1}{\cos(a^n) - 1} \right) \times \lim_{a \rightarrow 0} \left(\frac{\cos(a^n) - 1}{a^{2n}} \right) \times \lim_{a \rightarrow 0} a^{2n-m} \\ &= e \times 1 \times -\frac{1}{2} \times \lim_{a \rightarrow 0} a^{2n-m} \end{aligned}$$

Now $\lim_{a \rightarrow 0} a^{2n-m}$ must be equal to 1.

i.e. अर्थात्, $2n - m = 0$

$$\frac{m}{n} = 2$$

6. Let $a, b \in \mathbb{R}$ be such that $\lim_{x \rightarrow 0} \frac{x^2 \sin(\beta x)}{\alpha x - \sin x} = 1$. Then $6(\alpha + \beta)$ equals :

[JEE(Adv.)2016]

माना कि $a, b \in \mathbb{R}$ इस प्रकार हैं कि $\lim_{x \rightarrow 0} \frac{x^2 \sin(\beta x)}{\alpha x - \sin x} = 1$ है । तब $6(\alpha + \beta)$ का मान है :

[JEE(Adv.)2016]

Ans. 7

Sol.

$$\lim_{x \rightarrow 0} \frac{x^2 \left(\frac{\sin(\beta x)}{\beta x} \right) \beta x}{\alpha x - \sin x} = 1$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\beta x^3}{\alpha x - \left(x - \frac{x^3}{3} + \frac{x^5}{5} + \dots \right)} = 1 \quad \left(\because \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right)$$

$$\text{For finite limit} = \alpha - 1 = 0 \Rightarrow \alpha = 1$$

$$\text{Limit } 6\beta = 1, \beta = \frac{1}{6}$$

$$\Rightarrow 6(\alpha + \beta) = 6\left(1 + \frac{1}{6}\right) = 7$$

7. Let $f(x) = \frac{1-x(1+|1-x|)}{|1-x|} \cos\left(\frac{1}{1-x}\right)$ for $x \neq 1$. Then :

[JEE(Adv.)2017]

(A) $\lim_{x \rightarrow 1^+} f(x) = 0$

(B) $\lim_{x \rightarrow 1^-} f(x) = 0$

(C) $\lim_{x \rightarrow 1^+} f(x)$ does not exist

(D) $\lim_{x \rightarrow 1^-} f(x)$ does not exist

माना कि $x \neq 1$ के लिये, $f(x) = \frac{1-x(1+|1-x|)}{|1-x|} \cos\left(\frac{1}{1-x}\right)$ । तब :

[JEE(Adv.)2017]

(A) $\lim_{x \rightarrow 1^+} f(x) = 0$

(B) $\lim_{x \rightarrow 1^-} f(x) = 0$

(C) $\lim_{x \rightarrow 1^+} f(x)$ का अस्तित्व नहीं है

(D) $\lim_{x \rightarrow 1^-} f(x)$ का अस्तित्व नहीं है

Ans. BC

Sol. $f(1^+) = \lim_{h \rightarrow 0} \frac{1-(1+h)(1+h)}{h} \cos \frac{1}{h}$

$$= \lim_{h \rightarrow 0} \frac{1-(1+h)^2}{h} \cos \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-h^2 - 2h}{h} \cos \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} (-h - 2) \cos \frac{1}{h}$$

$$\Rightarrow \lim_{h \rightarrow 0} (1^+) \text{ does not exist}$$

$$f(1^-) \lim_{h \rightarrow 0} \frac{1-(1-h)(1+h)}{h} \cos \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2}{h} \cos \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} h \cos \frac{1}{h} = 0$$

8. Let e denote the base of the natural logarithm. The value of the real number a for which the right hand limit

$$\lim_{x \rightarrow 0^+} \frac{(1-x)^{\frac{1}{x}} - e^{-1}}{x^a} \text{ is equal to a nonzero real number, is } \underline{\hspace{2cm}}. \quad [\text{JEE(Adv.)2020}]$$

माना कि e प्राकृतिक लघुगणक के आधार को दर्शाता है। वास्तविक संख्या a का वो मान जिसके लिए दायें पक्ष की सीमा

$$\lim_{x \rightarrow 0^+} \frac{(1-x)^{\frac{1}{x}} - e^{-1}}{x^a} \text{ एक शून्येतर वास्तविक संख्या के बराबर है, है } \underline{\hspace{2cm}} \quad [\text{JEE(Adv.)2020}]$$

Ans. 1

Sol. $\lim_{x \rightarrow 0^+} \frac{(1-x)^{\frac{1}{x}} - e^{-1}}{x^a} \neq 0$

$$\lim_{x \rightarrow 0^+} \frac{e^{\frac{1}{x} \ln(1-x)} - e^{-1}}{x^a} \neq 0$$

$$\lim_{x \rightarrow 0^+} \frac{e^{-1} (e^{\frac{1}{x} \ln(1-x)} - 1)}{x^a} \neq 0$$

$$\lim_{x \rightarrow 0^+} \frac{1 + \frac{1}{x} \ln(1-x)}{e x^a} \neq 0$$

$$\lim_{x \rightarrow 0^+} \frac{x + \ln(1-x)}{e x^{a+1}} \neq 0$$

$$\lim_{x \rightarrow 0^+} \frac{x + \left(-x - \frac{x^2}{2} \dots \right)}{e x^{a+1}} \neq 0$$

$$\Rightarrow a+1 = 2$$

$$\Rightarrow a = 1$$

$$\text{if } a < 1 \Rightarrow \lim_{x \rightarrow 0^+} \frac{x + \left(-x - \frac{x^2}{2} \dots \right)}{e x^{a+1}} = 0$$

if $a > 1$ $\lim_{x \rightarrow 0^+} \frac{x + \left(-x - \frac{x^2}{2} \dots\right)}{e^{x^{a+1}}} = \text{DNE}$

9. The value of the limit $\lim_{x \rightarrow \frac{\pi}{2}} \frac{4\sqrt{2}(\sin 3x + \sin x)}{\left(2 \sin 2x \sin \frac{3x}{2} + \cos \frac{5x}{2}\right) - \left(\sqrt{2} + \sqrt{2} \cos 2x + \cos \frac{3x}{2}\right)} :$ [JEE(Adv.)2020]

सीमा $\lim_{x \rightarrow \frac{\pi}{2}} \frac{4\sqrt{2}(\sin 3x + \sin x)}{\left(2 \sin 2x \sin \frac{3x}{2} + \cos \frac{5x}{2}\right) - \left(\sqrt{2} + \sqrt{2} \cos 2x + \cos \frac{3x}{2}\right)}$ का मान है _____ [JEE(Adv.)2020]

Ans. 8

Sol. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{4\sqrt{2}(\sin 3x + \sin x)}{\left(2 \sin 2x \sin \frac{3x}{2} + \cos \frac{5x}{2}\right) - \left(\sqrt{2} + \sqrt{2} \cos 2x + \cos \frac{3x}{2}\right)}$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{4\sqrt{2}(2 \sin 2x \cos x)}{\left[\cos \frac{x}{2} - \cos \frac{7x}{2} + \cos \frac{5x}{2}\right] - \left[\sqrt{2}(2 \cos^2 x) + \cos \frac{3x}{2}\right]}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{4\sqrt{2}(2 \sin 2x \cos x)}{\cos \frac{x}{2} - \cos \frac{3x}{2} + \cos \frac{5x}{2} - \cos \frac{7x}{2} - 2\sqrt{2} \cos^2 x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{16\sqrt{2} \sin x \cos^2 x}{2 \sin x \sin \frac{x}{2} + 2 \sin 3x \sin \frac{x}{2} - 2\sqrt{2} \cos^2 x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{16\sqrt{2} \sin x \cos^2 x}{2 \sin \frac{x}{2} (\sin 3x + \sin x) - 2\sqrt{2} \cos^2 x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{16\sqrt{2} \sin x \cos^2 x}{2 \sin \frac{x}{2} (2 \sin 2x \cos x) - 2\sqrt{2} \cos^2 x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{16\sqrt{2} \sin x \cos^2 x}{8 \sin \frac{x}{2} \sin x \cos^2 x - 2\sqrt{2} \cos^2 x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{16\sqrt{2}}{8 \times \frac{1}{\sqrt{2}} - 2\sqrt{2}} = 8$$

10. For any positive integer n , let $S_n : (0, \infty) \rightarrow \mathbb{R}$ be defined by $S_n(x) = \sum_{k=1}^n \cot^{-1}\left(\frac{1+k(k+1)x^2}{x}\right)$, where for any $x \in \mathbb{R}$, $\cot^{-1}(x) \in (0, \pi)$ and $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then which of the following statements is (are) TRUE?

[JEE(Adv.)2021]

(A) $S_{10}(x) = \frac{\pi}{2} - \tan^{-1}\left(\frac{1+11x^2}{10x}\right)$, for all $x > 0$

(B) $\lim_{n \rightarrow \infty} \cot(S_n(x)) = x$, for all $x > 0$

(C) The equation $S_3(x) = \frac{\pi}{4}$ has a root in $(0, \infty)$

(D) $\tan(S_n(x)) \leq \frac{1}{2}$, for all $n \geq 1$ and $x > 0$

किसी धनात्मक पूर्णांक n के लिए, माना $S_n : (0, \infty) \rightarrow \mathbb{R}$, $S_n(x) = \sum_{k=1}^n \cot^{-1}\left(\frac{1+k(k+1)x^2}{x}\right)$ द्वारा परिभाषित है जहाँ किसी

$x \in \mathbb{R}$ के लिए $\cot^{-1}(x) \in (0, \pi)$ तथा $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ तब निम्न में से कौनसा/कौनसे कथन सही हैं/है?

[JEE(Adv.)2021]

(A) सभी $x > 0$ के लिए, $S_{10}(x) = \frac{\pi}{2} - \tan^{-1}\left(\frac{1+11x^2}{10x}\right)$

(B) सभी $x > 0$ के लिए $\lim_{n \rightarrow \infty} \cot(S_n(x)) = x$

(C) समीकरण $S_3(x) = \frac{\pi}{4}$ का एक मूल $(0, \infty)$ में है

(D) $\tan(S_n(x)) \leq \frac{1}{2}$, सभी $n \geq 1$ तथा $x > 0$ के लिए

Ans. AB

Sol.
$$S_n(x) = \sum_{k=1}^n \cot^{-1} \left(\frac{1+k(k+1)x^2}{x} \right)$$

$$= \sum_{k=1}^n \tan^{-1} \left(\frac{(k+1)x - kx}{1+(k+1)kx} \right)$$

$$= \sum_{k=1}^n \tan^{-1}(k+1)x - \tan^{-1} kx$$

$$= (\tan^{-1} 2x - \tan^{-1} x) + (\tan^{-1} 3x - \tan^{-1} 2x) + \dots + (\tan^{-1}(n+1)x - \tan^{-1} nx)$$

$$S_n(x) = \tan^{-1}(n+1)x - \tan^{-1} x$$

(A) $S_{10}(x) = \tan^{-1} 11x - \tan^{-1} x, x > 0$

$$= \tan^{-1} \left(\frac{11x - x}{1+11x.x} \right)$$

$$= \tan^{-1} \left(\frac{10x}{1+11x^2} \right)$$

$$= \cot^{-1} \left(\frac{1+11x^2}{10x} \right)$$

$$= \frac{\pi}{2} - \tan^{-1} \left(\frac{1+11x^2}{10x} \right) \Rightarrow \text{(A) is correct}$$

(B) $\lim_{n \rightarrow \infty} \cot(S_n(x)) = \lim_{n \rightarrow \infty} \cot(\tan^{-1}(n+1)x - \tan^{-1} x), x > 0$

$$= \lim_{n \rightarrow \infty} \cot \left(\tan^{-1} \frac{(n+1)x - x}{1+(n+1)x.x} \right)$$

$$= \lim_{n \rightarrow \infty} \cot \left(\tan^{-1} \frac{nx}{1+(n+1)x^2} \right)$$

$$= \lim_{n \rightarrow \infty} \cot \left[\tan^{-1} \frac{x}{\frac{1}{n} + \left(1 + \frac{1}{n}\right)x^2} \right] = \cot \tan^{-1} \left(\frac{x}{x^2} \right)$$

$$= \cot (\cot^{-1} x) = x$$

$\Rightarrow$ (B) is correct

(C) $S_3(x) = \frac{\pi}{4}$ has a root in $(0, \infty)$

$$\tan^{-1} 4x - \tan^{-1} x = \frac{\pi}{4}$$

$$\Rightarrow \frac{3x}{1+4x^2} = 1 \Rightarrow 4x^2 + 1 = 3x \Rightarrow 4x^2 - 3x + 1 = 0$$

$$D < 0$$

$\Rightarrow$ no real

$\Rightarrow$ (C) option is incorrect

$$(D) S_n(x) = \tan^{-1}(n+1)x - \tan^{-1}x$$

$$= \tan^{-1}\left(\frac{nx + x - x}{1 + (n+1)x^2}\right)$$

$$= \tan^{-1}\left(\frac{nx}{1 + (n+1)x^2}\right)$$

$$\tan(S_n(x)) = \frac{nx}{1 + (n+1)x^2} = \frac{n}{\frac{1}{x} + (n+1)x}$$

$$\Rightarrow A.M. \geq G.M.$$

$$\frac{\frac{1}{x} + (n+1)x}{2} \geq \sqrt{\frac{1}{x} \cdot (n+1)x}$$

$$\frac{1}{x} + (n+1)x \geq 2\sqrt{n+1}$$

$$\Rightarrow \tan(S_n(x)) = \frac{nx}{1 + (n+1)x^2} \leq \frac{n}{2\sqrt{n+1}}$$

for $n = 2$

$$\frac{n}{2\sqrt{n+1}} = \frac{1}{\sqrt{3}} > \frac{1}{2}$$

D is incorrect

11. If $\beta = \lim_{x \rightarrow 0} \frac{e^{x^3} - (1-x^3)^{\frac{1}{3}} + \left((1-x^2)^{\frac{1}{2}} - 1\right) \sin x}{x \sin^2 x}$ then the value of 6β is _____. [JEE(Adv.)2022]

यदि $\beta = \lim_{x \rightarrow 0} \frac{e^{x^3} - (1-x^3)^{\frac{1}{3}} + \left((1-x^2)^{\frac{1}{2}} - 1\right) \sin x}{x \sin^2 x}$, तब 6β का मान ज्ञात कीजिए। [JEE(Adv.)2022]

Ans. 5

Sol.
$$\beta = \lim_{x \rightarrow 0} \frac{e^{x^3} - (1 - x^3)^{1/3}}{\frac{x \sin^2 x}{x^2} x^2} + \frac{\left((1 - x^2)^{1/2} - 1 \right) \sin x}{x \frac{\sin^2 x}{x^2} x^2}$$

use expansion

$$\beta = \lim_{x \rightarrow 0} \frac{(1 + x^3) - \left(1 - \frac{x^3}{3}\right)}{x^3} + \lim_{x \rightarrow 0} \frac{\left(\left(1 - \frac{x^2}{2}\right) - 1\right) \sin x}{x^2 \cdot x}$$

$$\beta = \lim_{x \rightarrow 0} \frac{4x^3}{3x^3} + \lim_{x \rightarrow 0} \frac{-x^2}{2x^2}$$

$$\beta = \frac{4}{3} - \frac{1}{2} = \frac{5}{6}$$

$$6\beta = 5$$

12. Let $k \in \mathbb{R}$. If $\lim_{x \rightarrow 0^+} (\sin(\sin kx) + \cos x + x)^{\frac{2}{x}} = e^6$, then the value of k is : [JEE(Adv.)2024]

माना कि $k \in \mathbb{R}$ है। यदि $\lim_{x \rightarrow 0^+} (\sin(\sin kx) + \cos x + x)^{\frac{2}{x}} = e^6$, तब k का मान है : [JEE(Adv.)2024]

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol.
$$\lim_{x \rightarrow 0^+} (\sin(\sin kx) + \cos x + x)^{\frac{2}{x}} = e^6$$

$$e^{\lim_{x \rightarrow 0^+} 2 \left\{ \frac{\sin(\sin kx) + \cos x + x - 1}{x} \right\}} = e^6$$

$$\lim_{x \rightarrow 0^+} \frac{\sin(\sin kx) + \cos x + x - 1}{x} = 3$$

$$\lim_{x \rightarrow 0^+} \frac{\cos(\sin kx) \cdot \cos(kx) \cdot k - \sin x + 1}{1} = 3$$

$$k + 1 = 3 \Rightarrow k = 2$$

13. Let S be the set of all $(\alpha, \beta) \in \mathbb{R} \times \mathbb{R}$ such that $\lim_{x \rightarrow \infty} \frac{\sin(x^2)(\log_e x)^\alpha \sin\left(\frac{1}{x^2}\right)}{x^{\alpha\beta} (\log_e(1+x))^\beta} = 0$. Then which of the following

is (are) correct?

[JEE(Adv.)2024]

माना कि S उन सभी $(\alpha, \beta) \in \mathbb{R} \times \mathbb{R}$ का समुच्चय है कि $\lim_{x \rightarrow \infty} \frac{\sin(x^2)(\log_e x)^\alpha \sin\left(\frac{1}{x^2}\right)}{x^{\alpha\beta}(\log_e(1+x))^\beta} = 0$ है। तब निम्नलिखित में से

कौनसा(से) सही है (हैं)?

[JEE(Adv.)2024]

- (A) $(-1, 3) \in S$ (B) $(-1, 1) \in S$ (C) $(1, -1) \in S$ (D) $(1, -2) \in S$

Ans. BC

Sol.
$$\lim_{x \rightarrow \infty} \frac{\sin(x^2)(\ln(x))^\alpha \sin\left(\frac{1}{x^2}\right)}{x^{\alpha\beta}(\ln(1+x))^\beta} = 0$$

$$\lim_{x \rightarrow \infty} \frac{\sin(x^2) \ln(x)^\alpha \left(\frac{1}{x^2}\right)}{x^{\alpha\beta}(\ln(1+x))^\beta} = 0$$

$$\lim_{x \rightarrow \infty} \frac{\sin(x^2)(\ln x)^\alpha}{x^{\alpha\beta+2}(\ln(1+x))^\beta} = 0$$

$$\Rightarrow \alpha\beta + 2 > 0 \quad \text{or} \quad \alpha\beta + 2 = 0 \ \& \ \beta > \alpha$$

$$\alpha\beta > -2 \quad \alpha\beta = -2 \ \& \ \beta > \alpha$$

(A) $\alpha\beta = -3$ Incorrect

(B) $\alpha\beta = -1$ Correct

(C) $\alpha\beta = -1$ Correct

(D) $\alpha\beta = -2$ but $\alpha > \beta$ Incorrect

14. Let x_0 be the real number such that $e^{x_0} + x_0 = 0$. For a given real number α , define

$$g(x) = \frac{3xe^x + 3x - \alpha e^x - \alpha x}{3(e^x + 1)}$$

for all real numbers x .

Then which one of the following statements is TRUE?

[JEE(Adv.)2025]

(A) For $\alpha = 2$, $\lim_{x \rightarrow x_0} \left| \frac{g(x) + e^{x_0}}{x - x_0} \right| = 0$

(B) For $\alpha = 2$, $\lim_{x \rightarrow x_0} \left| \frac{g(x) + e^{x_0}}{x - x_0} \right| = 1$

(C) For $\alpha = 3$, $\lim_{x \rightarrow x_0} \left| \frac{g(x) + e^{x_0}}{x - x_0} \right| = 0$

(D) For $\alpha = 3$, $\lim_{x \rightarrow x_0} \left| \frac{g(x) + e^{x_0}}{x - x_0} \right| = \frac{2}{3}$

मान लीजिए कि x_0 वह वास्तविक संख्या (real number) है कि $e^{x_0} + x_0 = 0$ है। किसी दी गयी एक वास्तविक संख्या α , तथा सभी

वास्तविक संख्याओं x के लिए $g(x) = \frac{3xe^x + 3x - \alpha e^x - \alpha x}{3(e^x + 1)}$ परिभाषित कीजिये।

तब निम्नलिखित कथनों में से कौन सा एक कथन सत्य है?

[JEE(Adv.)2025]

(A) $\alpha = 2$ के लिए, $\lim_{x \rightarrow x_0} \left| \frac{g(x) + e^{x_0}}{x - x_0} \right| = 0$ है

(B) $\alpha = 2$ के लिए, $\lim_{x \rightarrow x_0} \left| \frac{g(x) + e^{x_0}}{x - x_0} \right| = 1$ है

(C) $\alpha = 3$ के लिए, $\lim_{x \rightarrow x_0} \left| \frac{g(x) + e^{x_0}}{x - x_0} \right| = 0$ है

(D) $\alpha = 3$ के लिए, $\lim_{x \rightarrow x_0} \left| \frac{g(x) + e^{x_0}}{x - x_0} \right| = \frac{2}{3}$ है

Ans. C

Sol. $g(x) = x - \frac{\alpha(e^x + x)}{3(e^x + 1)}$

$$g'(x) = 1 - \frac{\alpha}{3} \left\{ \frac{(e^x + 1)^2 - (e^x + x) \cdot e^x}{(e^x + 1)^2} \right\}$$

$$g'(x_0) = 1 - \frac{\alpha}{3} \left\{ \frac{(e^{x_0} + 1)^2 - 0}{(e^{x_0} + 1)^2} \right\}$$

$$g'(x_0) = 1 - \frac{\alpha}{3}$$

$$\lim_{x \rightarrow x_0} \left| \frac{g(x) + e^{x_0}}{x - x_0} \right| = \left(\frac{0}{0} \right)$$

Using L'Hospital Rule

$$= \lim_{x \rightarrow x_0} \left| \frac{g'(x)}{1} \right| = |g'(x_0)|$$

$$= \left| 1 - \frac{\alpha}{3} \right|$$

for $\alpha = 3 \Rightarrow L = 0$

1. Let $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) + \sin\left(\frac{1}{x^2}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$, then $\lim_{x \rightarrow \infty} f(x)$ equals :

- (A) 0 (B) $-1/2$ (C) 1 (D) none of these

माना $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) + \sin\left(\frac{1}{x^2}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$, तो $\lim_{x \rightarrow \infty} f(x) =$

- (A) 0 (B) $-1/2$ (C) 1 (D) इनमें से कोई नहीं

Ans. C

Sol. $\lim_{x \rightarrow \infty} x \sin\left(\frac{1}{x}\right) + \sin\left(\frac{1}{x^2}\right)$

$$\lim_{x \rightarrow \infty} \frac{\sin \frac{1}{x}}{\frac{1}{x}} + \frac{1}{x^2} \frac{\sin \frac{1}{x^2}}{\frac{1}{x^2}}$$

$$\text{Let } \frac{1}{x} = t \Rightarrow x \rightarrow \infty, t \rightarrow 0$$

$$\lim_{t \rightarrow 0} \frac{\sin t}{t} + t^2 \frac{\sin t}{t} = 1 + 0 = 1$$

2. Let α, β be the roots of $ax^2 + bx + c = 0$, where $1 < \alpha < \beta$. Then $\lim_{x \rightarrow x_0} \frac{|ax^2 + bx + c|}{ax^2 + bx + c} = 1$ then which of the following statements is incorrect :

- (A) $a > 0$ and $x_0 < 1$ (B) $a > 0$ and $x_0 > \beta$
(C) $a < 0$ and $\alpha < x_0 < \beta$ (D) $a < 0$ and $x_0 < 1$

माना समीकरण $ax^2 + bx + c = 0$ के मूल α, β हैं, जहाँ $1 < \alpha < \beta$. यदि $\lim_{x \rightarrow x_0} \frac{|ax^2 + bx + c|}{ax^2 + bx + c} = 1$ हो, तो निम्न कथनों में से

असत्य कथन है :

- (A) $a > 0$ और $x_0 < 1$ (B) $a > 0$ और $x_0 > \beta$
(C) $a < 0$ और $\alpha < x_0 < \beta$ (D) $a < 0$ और $x_0 < 1$

Ans. D

Sol. α, β be the roots of the equation $ax^2 + bx + c = 0$
where $1 < \alpha < \beta$.

- (i) if $a > 0$

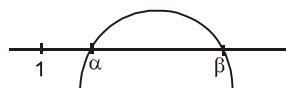


So $ax^2 + bx + c > 0$ when $x \in (-\infty, \alpha) \cup (\beta, \infty)$

$$\text{So } \lim_{x \rightarrow x_0} \frac{|ax^2 + bx + c|}{ax^2 + bx + c} = 1$$

when $a > 0$ and $x \in (-\infty, \alpha) \cup (\beta, \infty)$

(ii) If $a < 0$



So $ax^2 + bx + c > 0$ when $x \in (\alpha, \beta)$

$$\lim_{x \rightarrow x_0} \frac{|ax^2 + bx + c|}{ax^2 + bx + c} = 1$$

when $a < 0$ and $x \in (\alpha, \beta)$

So (A) $a > 0$ and $x_0 < 1$ right

(B) $a > 0$ and $x_0 > \beta$ right

(C) $a < 0$ and $\alpha < x_0 < \beta$ right

(D) $a < 0$ and $x_0 < 1$ wrong

3. If $\lim_{x \rightarrow 0} \frac{x^3}{\sqrt{a+x}(bx - \sin x)} = 1$ then the constants 'a' and 'b' are : (where $a > 0$)

यदि $\lim_{x \rightarrow 0} \frac{x^3}{\sqrt{a+x}(bx - \sin x)} = 1$ हो, तो a एवं b के मान हैं : (जहाँ $a > 0$)

(A) $b = 1, a = 36$

(B) $a = 1, b = 6$

(C) $a = 1, b = 36$

(D) $b = 1, a = 6$

Ans. A

Sol. $\lim_{x \rightarrow 0} \frac{x^3}{\sqrt{a+x}(bx - \sin x)} = 1$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x^3}{\sqrt{a}(bx - \sin x)} = 1$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{3x^2}{\sqrt{a}(b - \cos x)} = 1$$

$$\therefore b = 1$$

$$\therefore 1 = \lim_{x \rightarrow 0} \frac{3x^2}{\sqrt{a}(1 - \cos x)}$$

$$= \lim_{x \rightarrow 0} \frac{3x^2}{\sqrt{a} 2 \sin^2 \frac{x}{2}} = \frac{6}{\sqrt{a}}$$

$$\therefore a = 26$$

4. If $\ell = \lim_{x \rightarrow \infty} (\sin \sqrt{x+1} - \sin \sqrt{x})$ and $m = \lim_{x \rightarrow -\infty} [\sin \sqrt{x+1} - \sin \sqrt{x}]$ where $[.]$ denotes the greatest integer function then :

- (A) $\ell = m = 0$ (B) $\ell = 0$; m is undefined
(C) ℓ, m both do not exist (D) $\ell = 0, m \neq 0$ (although m exist)

यदि $\ell = \lim_{x \rightarrow \infty} (\sin \sqrt{x+1} - \sin \sqrt{x})$ और $m = \lim_{x \rightarrow -\infty} [\sin \sqrt{x+1} - \sin \sqrt{x}]$ है, जहाँ $[.]$ महत्तम पूर्णांक फलन को प्रदर्शित करता है, तो :

- (A) $\ell = m = 0$ (B) $\ell = 0$; m अपरिभाषित है
(C) ℓ, m दोनों विद्यमान नहीं है (D) $\ell = 0, m \neq 0$ (यद्यपि m विद्यमान है)

Ans. B

Sol. (i) $\ell = \lim_{x \rightarrow \infty} (\sin \sqrt{x+1} - \sin \sqrt{x})$

$$\Rightarrow \ell = \lim_{x \rightarrow \infty} 2 \cos \left(\frac{\sqrt{x+1} + \sqrt{x}}{2} \right) \cdot \sin \left(\frac{\sqrt{x+1} - \sqrt{x}}{2} \right)$$

$$\Rightarrow \ell = \lim_{x \rightarrow \infty} 2 \cos \left(\frac{\sqrt{x+1} + \sqrt{x}}{2} \right) \cdot \sin \left(\frac{x+1-x}{2(\sqrt{x+1} + \sqrt{x})} \right)$$

$$\Rightarrow \ell = \lim_{x \rightarrow \infty} 2 \cos \left(\frac{\sqrt{x} + \sqrt{x+1}}{2} \right) \sin \left(\frac{1}{2(\sqrt{x} + \sqrt{x+1})} \right)$$

$$\Rightarrow \ell = (\text{oscillating value } -1 \text{ to } 1) \times 0 = 0$$

(ii) $m = \lim_{x \rightarrow -\infty} [\sin \sqrt{x+1} - \sin \sqrt{x}]$

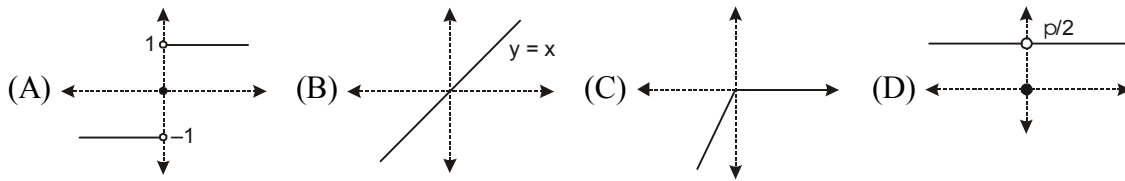
when $x \rightarrow -\infty$

then $\sqrt{x}$ undefined

m is undefined

5. The graph of the function $f(x) = \lim_{t \rightarrow 0} \left(\frac{2x}{\pi} \cot^{-1} \frac{x}{t^2} \right)$, is :

फलन $f(x) = \lim_{t \rightarrow 0} \left(\frac{2x}{\pi} \cot^{-1} \frac{x}{t^2} \right)$ का आरेख है :



Ans. C

Sol. $f(x) = \lim_{t \rightarrow 0} \left(\frac{2x}{\pi} \cot^{-1} \frac{x}{t^2} \right)$

Case-I : when $x = 0$

$$f(x) = 0$$

Case-II : when $x > 0$

$$f(x) = \lim_{t \rightarrow 0} \left(\frac{2x}{\pi} \cot^{-1} \frac{x}{t^2} \right) = \frac{2x}{\pi} \times \cot^{-1}(\infty) = \frac{2x}{\pi} \cdot 0 = 0$$

Case-III when $x < 0$

$$f(x) = \lim_{t \rightarrow 0} \left(\frac{2x}{\pi} \cot^{-1} \frac{x}{t^2} \right) = \frac{2x}{\pi} \times \cot^{-1}(-\infty) = \frac{2x}{\pi} \cdot \pi = 2x \Rightarrow f(x) = 2x$$

Hindi. $f(x) = \lim_{t \rightarrow 0} \left(\frac{2x}{\pi} \cot^{-1} \frac{x}{t^2} \right)$

स्थिति-I : जब $x = 0$

$$f(x) = 0$$

स्थिति-II : जब $x > 0$

$$f(x) = \lim_{t \rightarrow 0} \left(\frac{2x}{\pi} \cot^{-1} \frac{x}{t^2} \right) = \frac{2x}{\pi} \times \cot^{-1}(\infty) = \frac{2x}{\pi} \cdot 0 = 0$$

स्थिति-III जब $x < 0$

$$f(x) = \lim_{t \rightarrow 0} \left(\frac{2x}{\pi} \cot^{-1} \frac{x}{t^2} \right) = \frac{2x}{\pi} \times \cot^{-1}(-\infty) = \frac{2x}{\pi} \cdot \pi = 2x \Rightarrow f(x) = 2x$$

6. If $\lim_{x \rightarrow \infty} f(x)$ exist and is finite & nonzero and if $\lim_{x \rightarrow \infty} \left(f(x) + \frac{3f(x)-1}{f^2(x)} \right) = 3$ then the value of $\lim_{x \rightarrow \infty} f(x)$ is :

- (A) 1 (B) -1 (C) 2 (D) none of these

यदि $\lim_{x \rightarrow \infty} f(x)$ विद्यमान है तथा परिमित एवं अशून्य है और $\lim_{x \rightarrow \infty} \left(f(x) + \frac{3f(x)-1}{f^2(x)} \right) = 3$ हो, तो $\lim_{x \rightarrow \infty} f(x)$ का मान है :

- (A) 1 (B) -1 (C) 2 (D) इनमें से कोई नहीं

Ans. A

Sol. $\lim_{x \rightarrow \infty} \left(f(x) + \frac{3f(x)-1}{f^2(x)} \right) = 3$

Let $\lim_{x \rightarrow \infty} f(x) = y$

$$\text{or } \left(\lim_{x \rightarrow \infty} f(x) + \frac{3 \lim_{x \rightarrow \infty} f(x) - 1}{\left(\lim_{x \rightarrow \infty} f(x) \right)^2} \right) = 3$$

$$\text{or } \left(y + \frac{3y - 1}{y^2} \right) = 3$$

$$\text{or } y^3 - 3y^2 + 3y - 1 = 0$$

$$\text{or } (y - 1)^3 = 0$$

$$\text{or } y = 1$$

7. $\lim_{x \rightarrow \infty} \frac{e^x \left(\left(2^{x^n} \right)^{\frac{1}{e^x}} - \left(3^{x^n} \right)^{\frac{1}{e^x}} \right)}{x^n}$, $n \in \mathbb{N}$ is equal to :

(A) 0

(B) $\ln(2/3)$

(C) $\ln(3/2)$

(D) none

$$\lim_{x \rightarrow \infty} \frac{e^x \left(\left(2^{x^n} \right)^{\frac{1}{e^x}} - \left(3^{x^n} \right)^{\frac{1}{e^x}} \right)}{x^n}, n \in \mathbb{N} \text{ का मान है :}$$

(A) 0

(B) $\ln(2/3)$

(C) $\ln(3/2)$

(D) इनमें से कोई नहीं

Ans. B

Sol. $\lim_{x \rightarrow \infty} \frac{e^x \left(\left(2^{x^n} \right)^{\frac{1}{e^x}} - \left(3^{x^n} \right)^{\frac{1}{e^x}} \right)}{x^n}, n \in \mathbb{N}$

$$\lim_{x \rightarrow \infty} \frac{(2)^{\frac{x^n}{e^x}} - (3)^{\frac{x^n}{e^x}}}{\frac{x^n}{e^x}}$$

when $x \rightarrow \infty$, $\frac{x^n}{e^x} \rightarrow 0$

Put $\frac{x^n}{e^x} = t$

$$\lim_{t \rightarrow 0} \left(\frac{2^t - 3^t}{t} \right) = \ln 2 - \ln 3 = \ln \left(\frac{2}{3} \right)$$

8. If $\lim_{x \rightarrow 0} \frac{((a-n)x - \tan x) \sin nx}{x^2} = 0$, where n is non-zero real number, then a is equal to :

यदि $\lim_{x \rightarrow 0} \frac{((a-n)x - \tan x) \sin nx}{x^2} = 0$, जहाँ n अशून्य वास्तविक संख्या है, तो a का मान है :

- (A) 0 (B) $\frac{n+1}{n}$ (C) n (D) $n + \frac{1}{n}$

Ans. D

Sol. Given, $\lim_{x \rightarrow 0} \frac{\{(a-n)x - \tan x\} \sin nx}{x^2} = 0$

$$\Rightarrow \lim_{x \rightarrow 0} \left\{ (a-n)n - \frac{\tan x}{x} \right\} \frac{\sin nx}{nx} \times n = 0$$

$$\Rightarrow \{(a-n)n - 1\}n = 0$$

$$\Rightarrow (a-n)n = 1 \Rightarrow a = n + \frac{1}{n}$$

9. $\lim_{x \rightarrow 0} \frac{x \tan 2x - 2x \tan x}{(1 - \cos 2x)^2}$ is :

$\lim_{x \rightarrow 0} \frac{x \tan 2x - 2x \tan x}{(1 - \cos 2x)^2}$ का मान है :

- (A) 2 (B) -2 (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

Ans. C

Sol. $\lim_{x \rightarrow 0} \frac{x \tan 2x - 2x \tan x}{(1 - \cos 2x)^2}$ is

$$= \lim_{x \rightarrow 0} \frac{x \frac{2 \tan x}{1 - \tan^2 x} - 2x \tan x}{(2 \sin^2 x)^2}$$

$$= \lim_{x \rightarrow 0} \frac{2x \tan x \left[\frac{1}{1 - \tan^2 x} - 1 \right]}{4 \sin^4 x}$$

$$= \lim_{x \rightarrow 0} \frac{2x \tan x \left[\frac{1 - 1 + \tan^2 x}{1 - \tan^2 x} \right]}{4 \sin^4 x}$$

$$= \lim_{x \rightarrow 0} \frac{x \cdot \tan^3 x}{2 \sin^4 x (1 - \tan^2 x)}$$

$$= \lim_{x \rightarrow 0} \frac{1}{2} \cdot \frac{x \left(\frac{\tan x}{x} \right)^3 x^3}{\sin^4 x (1 - \tan^2 x)}$$

$$= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\left(\frac{\tan x}{x} \right)^3}{\left(\frac{\sin x}{x} \right)^4 (1 - \tan^2 x)}$$

$$= \frac{1(1)^3}{2(1)^4(1-0)} = \frac{1}{2}$$

10. The value of Limit $\frac{\cot^{-1}(x^{-a} \log_a x)}{\sec^{-1}(a^x \log_x a)}$ ($a > 1$) is equal to :

- (A) 1 (B) 0 (C) $\pi/2$ (D) does not exist

$$\lim_{x \rightarrow \infty} \frac{\cot^{-1}(x^{-a} \log_a x)}{\sec^{-1}(a^x \log_x a)} \quad (a > 1) \text{ का मान होगा :}$$

- (A) 1 (B) 0 (C) $\pi/2$ (D) अस्तित्वहीन

Ans. A

Sol. $\lim_{x \rightarrow \infty} \frac{\cot^{-1}\left(\frac{\log_a x}{x^a}\right)}{\sec^{-1}\left(\frac{a^x}{\log_a x}\right)} ; \text{ as } \left(\frac{\log_a x}{x^a}\right) \rightarrow 0 \text{ and } \left(\frac{a^x}{\log_a x}\right) \rightarrow \infty \text{ (using L'opital rule)}$

$$\therefore l = \frac{\pi/2}{\pi/2} = 1$$

11. $\lim_{x \rightarrow 0} \frac{\left(\sqrt{(1-\cos x) + \sqrt{(1-\cos x) + \sqrt{(1-\cos x) + \dots \infty}}} \right) - 1}{x^2}$ equals :

$$\lim_{x \rightarrow 0} \frac{\left(\sqrt{(1-\cos x) + \sqrt{(1-\cos x) + \sqrt{(1-\cos x) + \dots \infty}}} \right) - 1}{x^2} =$$

- (A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2

Ans. B

Sol. Let $\left(\sqrt{(1-\cos x)+\sqrt{(1-\cos x)+\sqrt{(1-\cos x)+\dots\infty}}}\right)=y$

$$y=\sqrt{(1-\cos x)+y}$$

$$\Rightarrow y^2-y+(\cos x-1)=0$$

$$y=\frac{1+\sqrt{5-4\cos x}}{2}$$

$$\begin{aligned}\text{Now } \lim_{x \rightarrow 0} \frac{\sqrt{5-4\cos x}-1}{2x^2} \\ = \lim_{x \rightarrow 0} \frac{4(1-\cos x)}{2x^2(\sqrt{5-4\cos x}+1)} = \frac{1}{2}\end{aligned}$$

12. $\lim_{x \rightarrow \infty} x \left(\tan^{-1} \frac{x+1}{x+2} - \cot^{-1} \frac{x+2}{x} \right)$ is

- (A) -1 (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) non existent

$\lim_{x \rightarrow \infty} x \left(\tan^{-1} \frac{x+1}{x+2} - \cot^{-1} \frac{x+2}{x} \right)$ का मान है -

- (A) -1 (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) विद्यमान नहीं है

Ans. B

Sol. For $x > 0 \Rightarrow \cot^{-1} \frac{x+2}{2} = \tan^{-1} \left(\frac{x}{x+2} \right)$

$$\lim_{x \rightarrow \infty} x \left(\tan^{-1} \left(\frac{x+1}{x+2} \right) - \tan^{-1} \left(\frac{x}{x+2} \right) \right)$$

$$= \lim_{x \rightarrow \infty} \left\{ x \left[\tan^{-1} \left(\frac{\frac{x+2}{x+2} - \frac{x}{x+2}}{1 + \left(\frac{x+1}{x+2} \times \frac{x}{x+2} \right)} \right) \right] \right\}$$

$$= \lim_{x \rightarrow \infty} \left\{ x \left[\tan^{-1} \left(\frac{(x+2)}{(x+2)^2 + x(x+1)} \right) \right] \right\}$$

$$\begin{aligned}
&= \lim_{x \rightarrow \infty} \left\{ x \left(\tan^{-1} \left(\frac{x+2}{2x^2+5x+4} \right) \right) \right\} \\
&= \lim_{x \rightarrow \infty} \left(\frac{\tan^{-1} \left(\frac{x+2}{2x^2+5x+4} \right)}{\left(\frac{x+2}{2x^2+5x+4} \right)} \right) \times \left(\frac{x(x+2)}{2x^2+5x+4} \right) \\
&= 1 \times \frac{1}{2} = \frac{1}{2}
\end{aligned}$$

13. Which one of the following statement is true?

(A) If $\lim_{x \rightarrow c} f(x) \cdot g(x)$ and $\lim_{x \rightarrow c} f(x)$ exist, then $\lim_{x \rightarrow c} g(x)$ exists.

(B) If $\lim_{x \rightarrow c} f(x) \cdot g(x)$ exists, then $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ exist.

(C) If $\lim_{x \rightarrow c} (f(x) + g(x))$ and $\lim_{x \rightarrow c} f(x)$ exist, then $\lim_{x \rightarrow c} g(x)$ exist.

(D) If $\lim_{x \rightarrow c} (f(x) + g(x))$ exists, then $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ exist

निम्न में से कौनसा कथन सत्य है ?

(A) यदि $\lim_{x \rightarrow c} f(x) \cdot g(x)$ तथा $\lim_{x \rightarrow c} f(x)$ विद्यमान है तो $\lim_{x \rightarrow c} g(x)$ विद्यमान होगी

(B) यदि $\lim_{x \rightarrow c} f(x) \cdot g(x)$ विद्यमान है तो $\lim_{x \rightarrow c} f(x)$ तथा $\lim_{x \rightarrow c} g(x)$ विद्यमान होगी

(C) यदि $\lim_{x \rightarrow c} (f(x) + g(x))$ तथा $\lim_{x \rightarrow c} f(x)$ विद्यमान है तो $\lim_{x \rightarrow c} g(x)$ विद्यमान होगी

(D) यदि $\lim_{x \rightarrow c} (f(x) + g(x))$ विद्यमान है तो $\lim_{x \rightarrow c} f(x)$ तथा $\lim_{x \rightarrow c} g(x)$ विद्यमान होगी

Ans. C

Sol. For option A and B, take $f(x) = x$ and $g(x) = \frac{1}{x}$.

Limit of $g(x)$ doesn't exist at $x = 0$, but for $\lim_{x \rightarrow 0} f(x) \cdot g(x)$ and $f(x) = x$ limit exists

For option C,

Let assume $h(x) = f(x) + g(x)$,

Take limit on both sides as $x \rightarrow c$

$\lim_{x \rightarrow c} h(x) = \lim_{x \rightarrow c} (f(x) + g(x))$

And it is given that $\lim_{x \rightarrow c} f(x)$ exists.

Using sum law of limits :

If $\lim_{x \rightarrow c} F(x)$ and $\lim_{x \rightarrow c} G(x)$ exists, then $\lim_{x \rightarrow c} (F(x) \pm G(x))$ also exists.

$\therefore \lim_{x \rightarrow c} (h(x) - f(x)) = \lim_{x \rightarrow c} (f(x) + g(x) - f(x)) = \lim_{x \rightarrow c} g(x)$ exists.

For option D, take $f(x) = -\frac{\cos x}{x^2}$ and $g(x) = \frac{1}{x^2}$.

Limit of $g(x)$ and $f(x) = -\frac{\cos x}{x^2}$ doesn't exist at $x=0$, but for $\lim_{x \rightarrow 0} f(x) + g(x)$ limit exists.

Hence option C.

14. If $f(x) = \cos x$, $x = n\pi$, $n = 0, 1, 2, 3, \dots$
 $= 3$, otherwise and

$$\phi(x) = \begin{cases} x^2 + 1 & \text{when } x \neq 3, x \neq 0 \\ 3 & \text{when } x = 0 \\ 5 & \text{when } x = 3 \end{cases} \quad \text{then } \lim_{x \rightarrow 0} f(\phi(x)) =$$

- (A) 1 (B) 3 (C) 5 (D) none

यदि $f(x) = \cos x$, $x = n\pi$, $n = 0, 1, 2, 3, \dots$

$= 3$ अन्यथा तथा

$$\phi(x) = \begin{cases} x^2 + 1 & \text{जब } x \neq 3, x \neq 0 \\ 3 & \text{जब } x = 0 \\ 5 & \text{जब } x = 3 \end{cases} \quad \text{है तो } \lim_{x \rightarrow 0} f(\phi(x)) =$$

- (A) 1 (B) 3 (C) 5 (D) कोई नहीं

Ans. B

Sol. $\lim_{x \rightarrow 0} \phi(x) = \lim_{x \rightarrow 0} (x^2 + 1)$
 $= 1$

$$\lim_{x \rightarrow 0} f(\phi(x)) = f(1) = 3$$

15. $\lim_{n \rightarrow \infty} \left(\frac{\sqrt[n]{p} + \sqrt[n]{q}}{2} \right)^n$, $p, q > 0$ equals :

$$\lim_{n \rightarrow \infty} \left(\frac{\sqrt[n]{p} + \sqrt[n]{q}}{2} \right)^n, \quad p, q > 0 \text{ का मान होगा :}$$

- (A) 1 (B) $\sqrt{pq}$ (C) pq (D) $\frac{pq}{2}$

Ans. B

Sol. $L = \lim_{n \rightarrow \infty} \left(\frac{p^{1/n} + q^{1/n}}{2} \right)^n = e^{\lim_{n \rightarrow \infty} n \left(\frac{p^{1/n} + q^{1/n}}{2} - 1 \right)} = e^{\lim_{n \rightarrow \infty} n \left(\frac{(p^{1/n} - 1) + (q^{1/n} - 1)}{2} \right)} = e^{\lim_{n \rightarrow \infty} \frac{1}{2} \left(\frac{p^{1/n} - 1}{1/n} + \frac{q^{1/n} - 1}{1/n} \right)}$

$$= e^{\left(\frac{\ln p + \ln q}{2}\right)} = e^{\left(\ln \sqrt{pq}\right)} = \sqrt{pq}$$

16. If $f(x) = \cot^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$ and $g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, then $\lim_{x \rightarrow a} \frac{f(x)-f(a)}{g(x)-g(a)}$, $0 < a < \frac{1}{2}$ is :

यदि $f(x) = \cot^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$ तथा $g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ है तो $\lim_{x \rightarrow a} \frac{f(x)-f(a)}{g(x)-g(a)}$, $0 < a < \frac{1}{2}$ का मान होगा :

(A) $\frac{3}{2(1+a^2)}$ (B) $\frac{3}{2}$ (C) $\frac{-3}{2(1+a^2)}$ (D) $-\frac{3}{2}$

Ans. D

Sol. $f(x) = \cot^{-1} \frac{3x-x^3}{1-3x^2} = \cot^{-1}(\tan 3\theta)$

$$\theta = \tan^{-1} x = \frac{\pi}{2} - 3 \tan^{-1} x$$

$$g(x) = \cos^{-1} \frac{1-x^2}{1+x^2} = 2 \tan^{-1} x$$

$$\lim_{x \rightarrow a} \frac{\frac{\pi}{2} - 3 \tan^{-1} x - \frac{\pi}{2} + 3 \tan^{-1} a}{2 \tan^{-1} x - 2 \tan^{-1} a}$$

$$= \frac{-3}{2} \times \frac{\tan^{-1} x - \tan^{-1} a}{\tan^{-1} x - \tan^{-1} a}$$

$$= \frac{-3}{2}$$

17. If $\lim_{x \rightarrow c^-} \{\ln x\}$ and $\lim_{x \rightarrow c^+} \{\ln x\}$ exists finitely but they are not equal (where $\{.\}$ denotes fractional part function), then :

- (A) 'c' can take only rational values
 (B) 'c' can take only irrational values
 (C) 'c' can take infinite values in which only one is irrational
 (D) 'c' can take infinite values in which only one is rational

यदि $\lim_{x \rightarrow c^-} \{\ln x\}$ तथा $\lim_{x \rightarrow c^+} \{\ln x\}$ परिमित रूप से विद्यमान है लेकिन दोनों का मान बराबर नहीं है (जहां $\{.\}$ भिन्नात्मक भाग फलन को दर्शाता है) तो :

- (A) 'c' केवल परिमेय संख्यायें ले सकता है
 (B) 'c' केवल अपरिमेय संख्याये ले सकता है
 (C) 'c' अनन्त संख्यायें ले सकता है जिनमें ठीक एक अपरिमेय हो
 (D) 'c' अनन्त संख्यायें ले सकता है जिनमें ठीक एक परिमेय हो

Ans. D

Sol. $\{\ln x\}$ is discontinuous at $\ln x \notin I$

$$x \notin e^I$$

$$x \neq e^I$$

18. Let $a = \lim_{x \rightarrow 1} \left(\frac{x}{\ln x} - \frac{1}{x \ln x} \right)$; $b = \lim_{x \rightarrow 0} \frac{x^3 - 16x}{4x + x^2}$; $c = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin x)}{x}$ and $d = \lim_{x \rightarrow -1} \frac{(x+1)^3}{3[\sin(x+1) - (x+1)]}$, then

the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is :

- (A) Idempotent (B) Involutary (C) Non-singular (D) Nilpotent

माना $a = \lim_{x \rightarrow 1} \left(\frac{x}{\ln x} - \frac{1}{x \ln x} \right)$; $b = \lim_{x \rightarrow 0} \frac{x^3 - 16x}{4x + x^2}$; $c = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin x)}{x}$ तथा $d = \lim_{x \rightarrow -1} \frac{(x+1)^3}{3[\sin(x+1) - (x+1)]}$ है तो

आव्यूह $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ होगा :

- (A) वर्गसम आव्यूह (B) अन्तर्वलनीय आव्यूह (C) व्युत्क्रमणीय आव्यूह (D) शून्यभावी आव्यूह

Ans. D

Sol. $a = \lim_{x \rightarrow 1} \left(\frac{x^2 - 1}{x \ln x} \right) = \lim_{x \rightarrow 1} \frac{x^2 - 1}{x(x-1)} = \lim_{x \rightarrow 1} \frac{x+1}{x}$

$$= 2$$

$$b = \lim_{x \rightarrow 0} \frac{x(x-4)(x+4)}{x(x+4)} = -4$$

$$c = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin x)}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$d = \lim_{x \rightarrow -1} \frac{(x+1)^3}{3 \left[(x+1) - \frac{(x+1)^3}{3!} + \dots - (x+1) \right]} = -2$$

$$\begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix} = 0$$

19. Consider the function $f(x) = \begin{cases} \frac{\max\left(x, \frac{1}{x}\right)}{\min\left(x, \frac{1}{x}\right)}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$, then $\lim_{x \rightarrow 0^-} \{f(x)\} + \lim_{x \rightarrow 1^-} \{f(x)\} + \lim_{x \rightarrow -1^-} [f(x)] =$

(where $\{.\}$ denotes fraction part function $[.]$ denotes greatest integer function)

माना फलन $f(x) = \begin{cases} \frac{\max\left(x, \frac{1}{x}\right)}{\min\left(x, \frac{1}{x}\right)}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$ है तो $\lim_{x \rightarrow 0^-} \{f(x)\} + \lim_{x \rightarrow 1^-} \{f(x)\} + \lim_{x \rightarrow -1^-} [f(x)] =$

(जहाँ $\{.\}$ भिन्नात्मक भाग फलन को दर्शाता है, $[.]$ महत्तम पूर्णांक फलन को दर्शाता है)

(A) 0

(B) 1

(C) 2

(D) 3

Ans. A

Sol. $f(x) = \begin{cases} \frac{1}{x^2} & x < -1 \\ x^2 & -1 < x < 0 \\ 1 & x = 0 \\ \frac{1}{x^2} & 0 < x < 1 \\ x^2 & x > 1 \end{cases}$

$$\lim_{x \rightarrow 0^-} \{f(x)\} + \lim_{x \rightarrow 1^-} \{f(x)\} + \lim_{x \rightarrow -1^-} [f(x)]$$

$$\lim_{x \rightarrow 0^-} \{x^2\} + \lim_{x \rightarrow 1^-} \left\{ \frac{1}{x^2} \right\} + \lim_{x \rightarrow -1^-} \left[\frac{1}{x^2} \right] = 0 + 0 + 0 = 0$$

20. Let $\alpha = \lim_{n \rightarrow \infty} \frac{(1^3 - 1^2) + (2^3 - 2^2) + \dots + (n^3 - n^2)}{n^4}$, then α is equal to :

(A) $\frac{1}{3}$

(B) $\frac{1}{4}$

(C) $\frac{1}{2}$

(D) non existent

माना $\alpha = \lim_{n \rightarrow \infty} \frac{(1^3 - 1^2) + (2^3 - 2^2) + \dots + (n^3 - n^2)}{n^4}$ है तो α का मान होगा :

- (A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) विद्यमान नहीं है

Ans. B

Sol. $\alpha = \lim_{n \rightarrow \infty} \frac{\sum n^3 - \sum n^2}{n^4}$

$$\alpha = \lim_{n \rightarrow \infty} \frac{\left(\frac{n[n+1]}{2}\right)^2 - \frac{n(n+1)(2n+1)}{6}}{n^4} = \frac{1}{4}$$

21. If $\lim_{x \rightarrow 0} [1 + x \ln(1+b^2)]^{\frac{1}{x}} = 2b \sin^2 \theta$, $b > 0$ and $\theta \in (-\pi, \pi]$, then the value of θ is : **[JEE-2011]**

यदि $\lim_{x \rightarrow 0} [1 + x \ln(1+b^2)]^{\frac{1}{x}} = 2b \sin^2 \theta$, $b > 0$ तथा $\theta \in (-\pi, \pi]$, तो θ का मान होगा : **[JEE-2011]**

- (A) $\pm \frac{\pi}{4}$ (B) $\pm \frac{\pi}{3}$ (C) $\pm \frac{\pi}{6}$ (D) $\pm \frac{\pi}{2}$

Ans. D

Sol. $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{(1-a)x^2 + x(1-a-b) + 1-b}{x+1} \right) = 4$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{\left((1-a).x + 1-a-b + \left(\frac{1-b}{x} \right) \right)}{1 + \frac{1}{x}} = 4$$

for limit to exist finitely

$$1 - a = 0 \text{ and } 1 - a - b = 4$$

$$\Rightarrow a = 1 \text{ and } b = -4.$$

ONE OR MORE THAN ONE CORRECT TYPE

22. Let $f(x) = \begin{cases} |x|, & -2 \leq x \leq 0 \\ e^x - 1, & 0 < x \leq 5 \end{cases}$, $g(x) = \begin{cases} -x-1, & -2 \leq x \leq 0 \\ \frac{4x-3}{3}, & 0 < x \leq 3 \end{cases}$ then :

माना $f(x) = \begin{cases} |x|, & -2 \leq x \leq 0 \\ e^x - 1, & 0 < x \leq 5 \end{cases}$, $g(x) = \begin{cases} -x-1, & -2 \leq x \leq 0 \\ \frac{4x-3}{3}, & 0 < x \leq 3 \end{cases}$ तो :

(A) $\lim_{x \rightarrow 0} f(g(x)) = 1$

(B) $\lim_{x \rightarrow 0^-} f(g(2^x)) = 1$

(C) $\lim_{x \rightarrow 0^+} f(g(2^x)) = e^{1/3} - 1$

(D) $\lim_{x \rightarrow 0^+} f(g(2^x - 1)) = \lim_{x \rightarrow 0^-} f(g(2^x - 1))$

Ans. ACD

Sol. (A) L.H.L. = $\lim_{x \rightarrow 0^-} f(g(x)) = \lim_{x \rightarrow 0^-} f(-x-1)$
 $= f(-1) = 1$

R.H.L. = $\lim_{x \rightarrow 0^+} f(g(x)) = \lim_{x \rightarrow 0^+} f\left(\frac{4x-3}{3}\right)$
 $= f(-1) = 1$

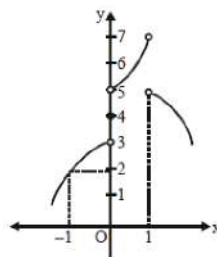
(B) & (C) $\lim_{x \rightarrow 0} f(g(2^x)) = \lim_{x \rightarrow 0} f\left(\frac{4 \cdot 2^x - 3}{3}\right)$
 $= f\left(\frac{1}{3}\right) = e^{1/3} - 1$

(D) $\lim_{x \rightarrow 0^+} f(g(2^x - 1)) = f(g(0^+)) = f(-1) = 1$

$\lim_{x \rightarrow 0^-} f(g(2^x - 1)) = f(g(0^-)) = f(-1) = 1$

23. The graph of $y = f(x)$ is as shown. The which of the following hold(s) good? [**Note** : $[k]$ and $\{k\}$ denote largest integer less than or equal k and fractional part of k respectively]

$y = f(x)$ का आरेख दिया गया है, तो निम्न में से कौनसे कथन सत्य हैं : (जहाँ $[.]$ तथा $\{.\}$ महत्तम पूर्णांक तथा भिन्नात्मक भाग फलन दर्शाता है)



$$(A) \lim_{x \rightarrow 0^+} f([x - \sin x]) = 4$$

$$(B) \lim_{x \rightarrow 0^-} f([x - \tan x]) = 2$$

$$(C) \lim_{x \rightarrow 0^+} f\left(\left\{\frac{x}{\tan x}\right\}\right) = 7$$

$$(D) \lim_{x \rightarrow \frac{\pi}{2}^+} f(\{\cos x\}) = 5$$

Ans. AC

Sol. (A) $x > \sin x$

$$\lim_{x \rightarrow 0^+} f([x - \sin x]) = \lim_{x \rightarrow 0^+} f(0) = 4$$

$$(B) x < 0 \Rightarrow x > \tan x \\ \Rightarrow x - \tan x > 0$$

$$\lim_{x \rightarrow 0} f([x - \tan x]) = \lim_{x \rightarrow 0^+} f(0) = 4$$

$$(C) \tan x > x$$

$$\Rightarrow \frac{x}{\tan x} < 1$$

$$\lim_{x \rightarrow 0^+} f\left\{\frac{x}{\tan x}\right\} = \lim_{x \rightarrow 0^+} f(1^-) = 7$$

$$(D) \lim_{x \rightarrow \frac{\pi}{2}^+} f\{\cos x\} = \lim_{x \rightarrow \frac{\pi}{2}^+} f\{-1\} = \lim_{x \rightarrow \frac{\pi}{2}^+} f(-1) = 7$$

24. Let $f(x) = \frac{\cos^{-1}(\cos x)}{x}$, then which of the following is (are) correct : (where $[.]$ denotes greatest integer function)

माना $f(x) = \frac{\cos^{-1}(\cos x)}{x}$, तो निम्न में से कौनसे कथन सत्य है : (जहाँ $[.]$ महत्तम पूर्णांक फलन दर्शाता है) ?

$$(A) \lim_{x \rightarrow 0^+} [f(x)] = 1$$

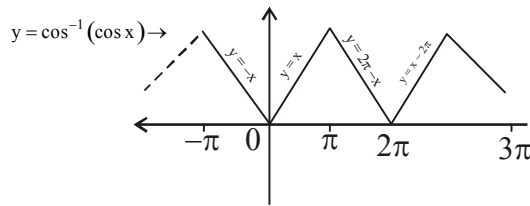
$$(B) \lim_{x \rightarrow 0^-} [f(x)] = 1$$

$$(C) \lim_{x \rightarrow 0^+} [f(x)] = 0$$

$$(D) \lim_{x \rightarrow 0^-} [f(x)] = -1$$

Ans. AD

Sol. $f(x) = \frac{\cos^{-1}(\cos x)}{x}$



So, clearly when $x \rightarrow 0^+$,

$$\rightarrow \lim_{x \rightarrow 0^+} \left[\frac{\cos^{-1}(\cos x)}{x} \right]$$

$$= \lim_{h \rightarrow 0} \left[\frac{\cos^{-1}(\cos(0+h))}{0+h} \right]$$

$$= \lim_{h \rightarrow 0} \left[\frac{h}{h} \right] = 1$$

$$= \lim_{x \rightarrow 0^-} \left[\frac{\cos^{-1}(\cos x)}{x} \right]$$

$$= \lim_{h \rightarrow 0} \left[\frac{\cos^{-1}(\cos(0-h))}{0-h} \right] = \lim_{h \rightarrow 0} \left[\frac{h}{-h} \right]$$

$$= -1$$

$\therefore$ Hence, Option (A), (D) are correct.

25. Which of the following limits does not exist? (where $[.]$ represents greatest integer function)

निम्न में से कौनसी सीमा विद्यमान नहीं होगी : (जहाँ $[.]$ महत्तम पूर्णांक फलन दर्शाता है)

(A) $\lim_{x \rightarrow 1^+} ([x])^{\frac{1}{x-1}}$

(B) $\lim_{x \rightarrow 3} \frac{(x^2 - 9 - \sqrt{x^2 - 6x + 9})}{|x-1| - 2}$

(C) $\lim_{x \rightarrow 0^+} (x)^{\ln x}$

(D) $\lim_{x \rightarrow 0^+} \left(\frac{1 - \cos(\sin^2 x)}{x^2} \right)^{\frac{\ln(1-2x^2)}{\sin^2 x}}$

Ans. BCD

Sol. (A) $\lim_{x \rightarrow 1^+} [x]^{\frac{1}{x-1}} = \lim_{x \rightarrow 1^+} 1^{\frac{1}{x-1}} = 1$

$$(B) \lim_{x \rightarrow 3} \frac{x^2 - 9\sqrt{(x-3)^2}}{|x-1|-2} = \lim_{x \rightarrow 3} \frac{(x-3)(x+3) - |x-3|}{(x-3)}$$

$$\text{L.H.L.} = \lim_{x \rightarrow 3^-} \frac{(x-3)(x+3) + (x-3)}{(x-3)} = \lim_{x \rightarrow 3^-} x + 4 = 7$$

$$\text{R.H.L.} = \lim_{x \rightarrow 3^+} \frac{(x-3)(x+3) - (x-3)}{(x-3)} = \lim_{x \rightarrow 3^+} x + 2 = 5$$

LHL $\neq$ RHL limit DNE

$$(C) \lim_{x \rightarrow 0^+} (x)^{\ln x} = \lim_{x \rightarrow 0^+} e^{(\ln x)^2} = e^\infty \neq \text{finite}$$

Limit DNE

$$(D) \lim_{x \rightarrow 0^+} \left(\frac{1 - \cos \sin^2 x}{x^2} \right)^{\frac{\ln(1-2x^2)}{\sin^2 x}} = \lim_{x \rightarrow 0^+} (f(x))^{g(x)}$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{1 - \cos \sin^2 x}{x^2} = \lim_{x \rightarrow 0^+} \frac{1 - \cos \sin^2 x}{\sin^4 x} \times \frac{\sin^2 x}{x^2} \times \sin^2 x = \frac{1}{2} \times 1 \times 0 = 0$$

$$\lim_{x \rightarrow 0^+} g(x) = \lim_{x \rightarrow 0^+} \frac{\ln(1-2x^2)}{\sin^2 x} = \lim_{x \rightarrow 0^+} \frac{-2x^2}{\sin^2 x} = -2$$

$$\lim_{x \rightarrow 0^+} (f(x))^{g(x)} = 0^{-2} \neq \text{Neither finite nor indeterminate form.}$$

26. Which of the following limits are non existent ?

$$(A) \lim_{x \rightarrow 0} \frac{1}{x} \sqrt{\frac{1 - \cos x}{1 + \cos x}} \quad (B) \lim_{x \rightarrow 0} \frac{|x|}{x} \quad (C) \lim_{x \rightarrow 0} \frac{\cot^{-1} \frac{1}{x}}{x}$$

$$(D) \lim_{x \rightarrow \pi/2} \cos^{-1}[\cot x] \text{ where } [] \text{ is greatest integer function}$$

निम्न में से कौनसी सीमा विद्यमान नहीं होगी :

$$(A) \lim_{x \rightarrow 0} \frac{1}{x} \sqrt{\frac{1 - \cos x}{1 + \cos x}} \quad (B) \lim_{x \rightarrow 0} \frac{|x|}{x} \quad (C) \lim_{x \rightarrow 0} \frac{\cot^{-1} \frac{1}{x}}{x}$$

$$(D) \lim_{x \rightarrow \pi/2} \cos^{-1}[\cot x] \text{ (जहाँ } [.] \text{ महत्तम पूर्णांक फलन दर्शाता है)}$$

Ans. ABCD

Sol. (A) $\lim_{x \rightarrow 0} \frac{|\tan x|}{x}$

$$\text{LHL} = -1, \text{RHL} = 1$$

$$(B) \lim_{x \rightarrow 0} \frac{|x|}{x} \Rightarrow \text{LHL} = -1, \text{RHL} = 1$$

$$(C) \lim_{x \rightarrow 0} \frac{\cot^{-1} \frac{1}{x}}{x}$$

$$\text{LHL} = \lim_{x \rightarrow 0^-} \frac{\pi + \tan^{-1} x}{x} \neq \text{finite (DNE)}$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} \frac{\tan^{-1} x}{x} = 1$$

$$(D) \lim_{x \rightarrow 0} \cos^{-1} [\cot x]$$

$$\text{LHL} = \lim_{x \rightarrow 0^-} \cos^{-1} (-1) = \pi$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} \cos^{-1} (0) = 0$$

$$27. \text{ Let } f(x) = \begin{cases} \frac{x}{\tan x} & , -\infty < x < 0 \\ 2 & , x = 0 \\ \frac{\ln(1+2x)}{x} & , 0 < x < \infty \end{cases} \text{ and } g(x) = \begin{cases} x+4 & , -\infty < x < 1 \\ x^2-5x+11 & , 1 \leq x < 2 \\ x-3 & , 2 \leq x < \infty \end{cases} \text{ , then which of the following}$$

is(are) correct?

$$\text{माना } f(x) = \begin{cases} \frac{x}{\tan x} & , -\infty < x < 0 \\ 2 & , x = 0 \\ \frac{\ln(1+2x)}{x} & , 0 < x < \infty \end{cases} \text{ तथा } g(x) = \begin{cases} x+4 & , -\infty < x < 1 \\ x^2-5x+11 & , 1 \leq x < 2 \\ x-3 & , 2 \leq x < \infty \end{cases} \text{ , तो निम्न में से कौनसे कथन}$$

सत्य हैं :

$$(A) \lim_{x \rightarrow 0^-} g(f(x)) = 5$$

$$(B) \lim_{x \rightarrow 0^-} g(f(x)) = 7$$

$$(C) \lim_{x \rightarrow 0^+} g(f(x)) = 5$$

$$(D) \lim_{x \rightarrow 0^+} g(f(x)) = -1$$

Ans. AC

$$\text{Sol. } \frac{x}{\tan x} < 1$$

$$(A) \lim_{x \rightarrow 0^-} g(f(x)) = \lim_{x \rightarrow 0^-} g\left(\frac{x}{\tan x}\right) = \lim_{x \rightarrow 0^-} g(1^-) = 5$$

$$(C) \lim_{x \rightarrow 0^+} g(f(x)) = \lim_{x \rightarrow 0^+} g\left(\frac{x}{\tan x}\right) = \lim_{x \rightarrow 0^+} g(1^-) = 5$$

28. Let $f(x) = \begin{cases} |x-2| + a^2 - 6a + 9, & x < 2 \\ 5 - 2x, & x \geq 2 \end{cases}$

If $\lim_{x \rightarrow 2} [f(x)]$ exists, the possible values a can take is/are (where $[.]$ represents the greatest integer function)

$$\text{माना } f(x) = \begin{cases} |x-2| + a^2 - 6a + 9, & x < 2 \\ 5 - 2x, & x \geq 2 \end{cases}$$

यदि $\lim_{x \rightarrow 2} [f(x)]$ विद्यमान है, तो a के सम्भावित मान जो a ले सकता है (जहां $[.]$ महत्तम पूर्णांक फलन को दर्शाता है)

(A) 2 (B) $\frac{5}{2}$ (C) 3 (D) $\frac{7}{2}$

Ans. BCD

Sol. $LHL = \lim_{x \rightarrow 2^-} [|x-2| + (a-3)^2] = [(a-3)^2]$

$$RHL = \lim_{x \rightarrow 2^+} [5 - 2x] = 0$$

$$(a-3)^2 \in [0, 1) \Rightarrow a \in (2, 4)$$

29. Let $L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4}$, $a > 0$. If L is finite, then :

[JEE-2009]

माना कि $L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4}$, $a > 0$ यदि L एक परिमित संख्या हो तो सही विकल्प चुनो :

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(A) $a = 2$ (B) $a = 1$ (C) $L = \frac{1}{64}$ (D) $L = \frac{1}{32}$

Ans. AC

Sol.
$$\frac{a - a \left(1 - \frac{x^2}{a^2}\right)^{\frac{1}{2}} - \frac{x^2}{4}}{x^4} = \frac{a - a \left(1 - \frac{x^2}{2a^2} - \frac{1}{8} \frac{x^4}{a^4}\right) - \frac{x^2}{4}}{x^4}$$

$$a = 2, (\text{coefficient of } x^2 = 0)$$

$$\therefore L = \frac{1}{16}$$